Executive Summary

SALS Marketing Inc. was tasked with designing an advertising campaign that maximizes exposure while staying within the client's $800,000 budget. Using a mixed-integer linear programming model implemented in CVXPY, we identified the optimal media mix that maximizes total exposure while satisfying all business constraints and benchmarked it against a naive split baseline to quantify the value of optimization.

Bottom Line:

The optimized plan delivered **14,235,000 impressions at $799,000. 25.53% more reach than what an equal split baseline at the same budget can achieve**.

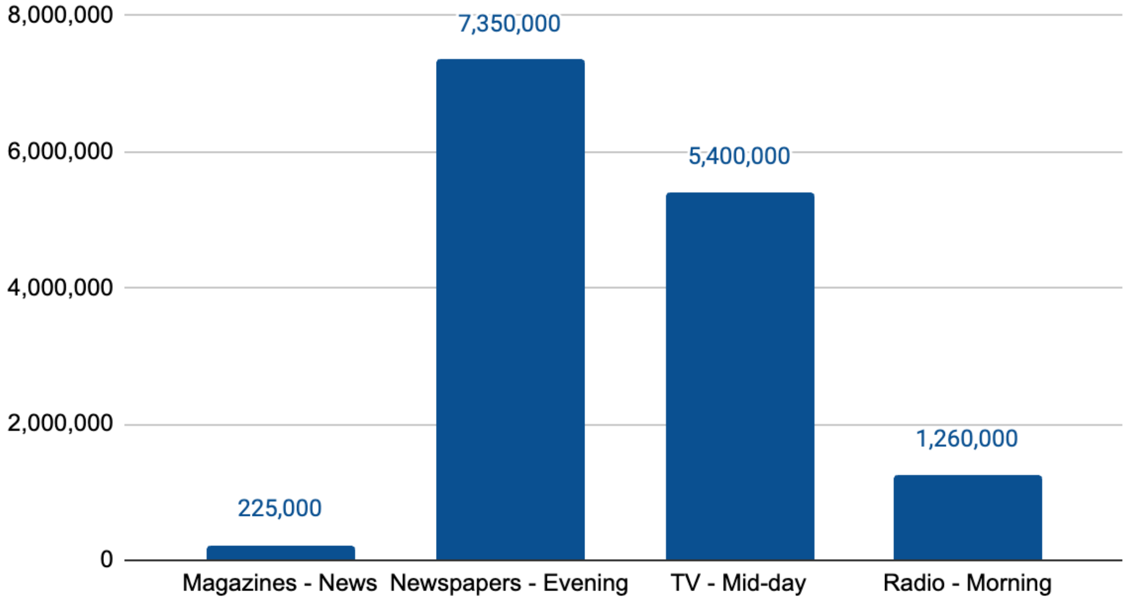
Optimal Media Mix

Channel	Spend	Impressions
Evenining Newspaper	$294,000	7,350,000
Mid-Day TV	$300,000	5,400,000
A.M. Radio	$105,000	1,260,000
News Magazines	$100,000	225,000
Total	***$799,000***	***14,235,000***

Recommendations

- Roll out this media mix for the current campaign
- Revisit the $100,000 category minimums as relaxing them could unlock additional revenue
- Rerun the model quarterly

Media Exposure Contribution



Percentage of Budget Allocation